Effect of Acetylacetone on the Properties of Nebulized Spray Deposited TiO_2 Films

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***Abstract*— With arousing interest in the preparation of anatase TiO_2 films in several green energy applications, various methods have emerged for the same in recent days. Herein, anatase TiO_2 films are conveniently obtained by nebulized spray deposition using isopropanol as a solvent, acetylacetone as a stabilizer and titanium tetraisopropoxide as a titanium source. The concentration of acetylacetone is varied as 1 ml, 2 ml, 3 ml and its influence on the properties of anatase TiO_2 films is investigated. The XRD and Raman studies indicated the formation of phase pure anatase films. FESEM images confirmed the smooth surface morphology of the obtained films. Broad band was observed in the visible region of photoluminescence spectra. Further, the UV-DRS study confirmed the reduced reflection properties of the TiO_2 films. The band gap energy of the obtained films varied from 3.0 eV to 2.95 eV as observed from Kubelka-Munk indirect band gap plot of TiO_2 films.**

Keywords: Thin films, TiO_2, Spray deposition

I. INTRODUCTION

Anatase TiO_2 thin films have been extensively investigated due to their outstanding properties such as better chemical and thermal stability, wide band gap and high dielectric constant [1]. These films have applications in wide variety of areas such as self cleaning applications, dye sensitized solar cells and supercapacitor [2-3]. Various methods such as spin coating, spray deposition, magnetron sputtering and hydrothermal have emerged for the preparation of TiO_2 films [1]. Among them nebulized spray deposition is simple and cost effective technique for the preparation of anatase TiO_2 thin films. In nebulized spray deposition various parameters such as precursor source, substrate temperature and solvent etc. can be changed to tune the properties of the resultant films. In the present work, the stabilizer acetylacetone concentration is varied and its effect on the properties of TiO_2 thin films is investigated.

II. EXPERIMENTAL

Prior to deposition the glass substrates were cleaned using hot chromic acid to remove the organic impurities on the surface of the films. The glass substrates were further cleaned with acetone subsequently treated ultrasonically for 15 minutes and then dried in air oven for an hour before deposition.

In the present work, anatase TiO_2 thin films are prepared by the following procedure. 1 ml of acetylacetone is added to the 8 ml of isopropanol solution, further 1 ml of titanium tetraisopropoxide is added to the same solution. The resultant solution is ultrasonicated for 20 minutes to get a uniform dispersion. The prepared solution was sprayed onto fluorine doped tin oxide (FTO) coated glass substrates kept inside the home made furnace at a substrate temperature of 340°C for 10 minutes. The as-deposited films were further annealed at 450°C for an hour to remove the organic residues from the films. The films were also prepared by varying the acetylacetone concentration as 2 ml and 3 ml and by keeping other conditions remain constant.

III. RESULTS AND DISCUSSION

Phase analysis

Figure 1 shows the Raman spectra of TiO2 thin films prepared at various concentrations of acetylacetone. The observed Raman spectra confirm that obtained TiO2 thin films corresponds to anatase phase [3]. The anatase exhibits a tetragonal structure, belongs to the space group D144h (I41/amd). Its conventional cell is made up of two primitive cells, each with two TiO2 units. The spectra has six Raman active anatase modes (A1g + 2B1g + 3Eg), which were observed at A1g (512 cm−1), 2B1g (393 cm−1 and 512 cm-1) and 3Eg (141 cm−1, 197 cm−1, and 633 cm−1) [3]. No other peak was observed related to rutile or brookite phase. With the increasing concentration of acetylacetone the Raman peak intensity is found to decrease, which revealed the high crystalline nature of the film prepared with 1 ml of acetylacetone..

Surface morphology

Figure 2 shows the surface morphology of anatase TiO_2 thin films prepared at different concentrations of acetylacetone. All the films show uniformly distributed crystal grains with very smooth surface. In addition, the size of grains is found to increase with increasing concentration of acetylacetone.

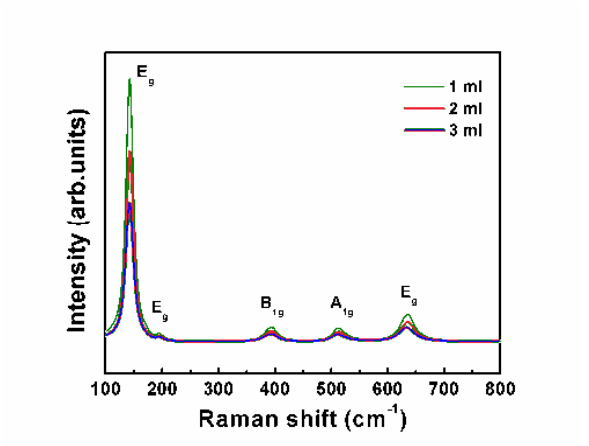

FIGURE 1. Raman spectra of anatase TiO_2 thin films prepared at different concentrations of acetylacetone.

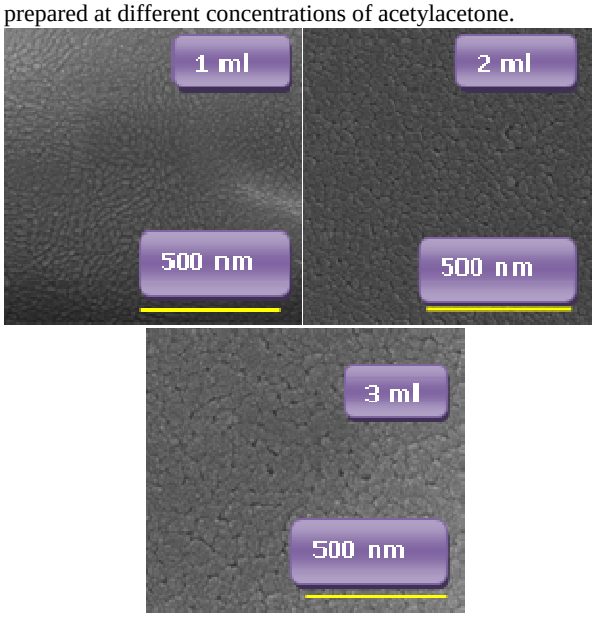

FIGURE 2. Surface morphology of anatase tio_2 thin films prepared at different concentrations of acetylacetone.

Photoluminescence

Figure 3 shows the photoluminescence spectra of anatase TiO_2 thin films. The spectra shows a broad band in the visible region which is usually observed in anatase phase TiO_2. The observed broad band is deconvoluted (Gaussian fitting) into four peaks (360, 385, 424 and 483 nm) as shown in inset. The 360 nm peak is corresponds to the near band edge emission, the band edge emission was observed at 385 nm and the peaks at 424, 483 nm can be assigned to shallow trap levels [2].

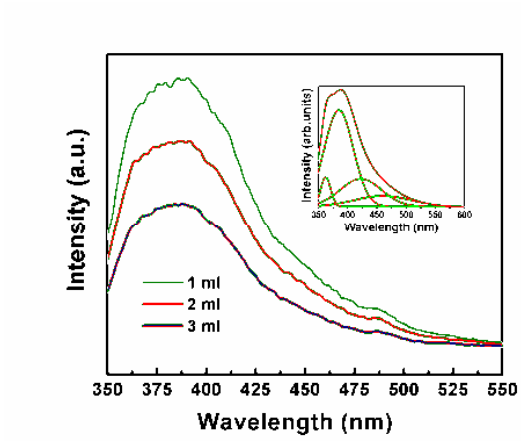

FIGURE 3. Photoluminescence spectra of anatase TiO_2 thin films prepared at different concentrations of acetylacetone.

Optical properties

Figure 4 shows the optical absorption spectra of anatase TiO_2 thin films prepared at different concentrations of acetylacetone. The observed films shows a cut off region around 395 nm which is due to the metal ligand charge transfer between Ti^{4+} and O^{2-} ions [3]. The observed spectra of the anatase TiO_2 thin films show a slight enhancement of absorption in the visible region.

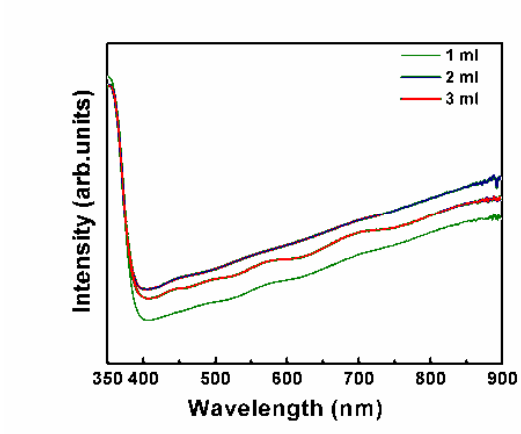

FIGURE 4. Optical absorption spectra of anatase TiO_2 thin films prepared at different concentrations of acetylacetone.

Figure 5 shows the reflectance spectra of anatase TiO_2 thin films prepared at different concentrations of acetylacetone. The obtained films show low reflectance values which are suitable for applications such as dye sensitized solar cells. The observed reflectance value for 1 ml films is 26%, 2 ml film is 23% and 3

ml film is 24%. The slight deviation in the reflectance values may be due to the variation in film thickness.

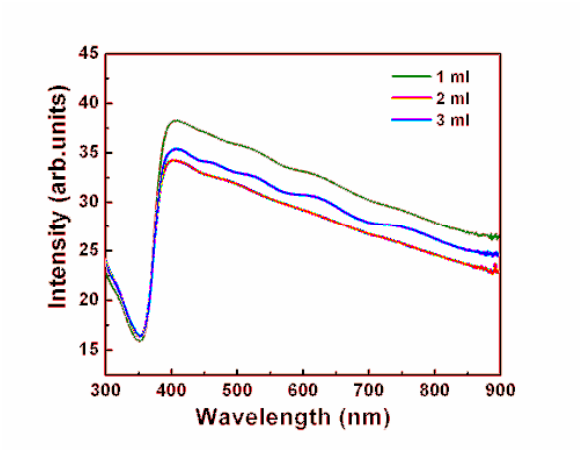

FIGURE 5. Reflectance spectra of anatase TiO_2 thin films prepared at different concentrations of acetylacetone.

Figure 6 shows the Kubelka-Munk indirect band gap plot of anatase TiO_2 thin films prepared at different concentrations of acetylacetone. Kubelka-Munk plot was used to calculate the band gap of TiO_2 films [3]. The indirect band gap values of the films were obtained by plotting a graph of $[F(R)h\nu]^{1/2}$ against hν as shown in Figure 6. The obtained anatase TiO_2 thin films have a band gap of 3.0 eV which matches with reported values. With increase in concentration of acetylacetone there is a slight decrease in band gap from 3.0 eV to 2.95 eV. The observed band gap reduction can be due to various reasons such as thickness, grain size, quantum effects and mass density.

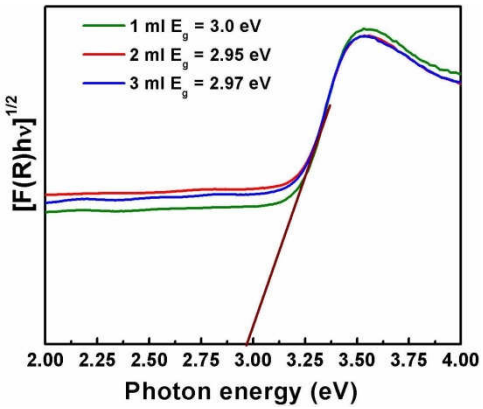

FIGURE 6. Kubelka-Munk indirect band gap plot of anatase TiO_2 thin films prepared at different concentrations of acetylacetone.

IV. CONCLUSION

Anatase TiO_2 thin films were successfully prepared by varying the concentrations of acetylacetone. The obtained films exhibit anatase phase. The films possess a smooth surface with homogeneous distribution grain as confirmed by SEM images. The Kubelka-Munk plot shows that obtained films band gap vary from 3.0 eV to 2.95 eV.

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